Olist Business Rules & Key Metrics Guide

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1. Overview

This document defines the key business metrics and rules used when analyzing the Olist e-commerce dataset.

It ensures:

- consistency in calculations
- correct interpretation of results
- alignment with business analysis best practices

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2. Core Metrics

Total Revenue (GMV – Gross Merchandise Value)

Definition:
Total value of products sold.

Calculation:

- Sum of order_items.price

Notes:

- Does NOT include freight (shipping) unless explicitly stated
- Calculated at the item level

Total Orders

Definition:
Total number of unique orders.

Calculation:

- Count of distinct order_id in orders

Average Order Value (AOV)

Definition:
Average value per order.

Calculation:

- Total Revenue ÷ Total Orders

Total Customers

Definition:
Number of unique customers.

Calculation:

- Count of distinct customer_unique_id

Repeat Purchase Rate

Definition:
Percentage of customers who made more than one purchase.

Calculation:

- Customers with >1 order ÷ total customers

Total Payments

Definition:
Total value of payments recorded.

Calculation:

- Sum of payments.payment_value

Note:

- May differ slightly from revenue due to installments or adjustments

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3. Delivery & Logistics Metrics

Delivery Time

Definition:
Time between purchase and delivery.

Calculation:

- order_delivered_customer_date - order_purchase_timestamp

Delivery Delay

Definition:
Difference between actual delivery date and estimated delivery date.

Calculation:

- order_delivered_customer_date - order_estimated_delivery_date

Interpretation:

- Positive → late delivery
- Negative → early delivery

Delivered Orders

Definition:
Orders successfully delivered to customers.

Filter:

- order_status = 'delivered'

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4. Customer Satisfaction Metrics

Average Review Score

Definition:
Average rating given by customers.

Calculation:

- Average of review_score

Low Review Rate

Definition:
Percentage of reviews with low ratings (1 or 2).

Calculation:

- Count of low reviews ÷ total reviews

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5. Product & Seller Metrics

Top Products

Definition:
Products with highest sales or revenue.

Common calculations:

- Count of orders per product
- Sum of revenue per product

Top Categories

Definition:
Product categories generating the most revenue or orders.

Seller Performance

Definition:
Evaluation of sellers based on:

- total sales
- number of orders
- average review score

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6. Important Business Rules

- Revenue should be calculated using order_items.price, not payments.payment_value
- Orders should be counted using **distinct order_id**
- Customer-level analysis should use customer_unique_id, not customer_id
- Delivery metrics should only consider **delivered orders**
- Join tables carefully:
    - orders ↔ order_items ↔ products
    - orders ↔ payments
    - orders ↔ reviews

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7. Notes for Analysis

- Always clarify:
    - time period (monthly, yearly, etc.)
    - geographic scope (state, city)
    - product category if relevant
- When results seem inconsistent:
    - check joins
    - verify aggregation level (order vs item)